

RUT Response to the April 2026 H0 Distance Network Result

GitHub / OSF version

Summary

The April 2026 H0 Distance Network result does **not** settle the Hubble constant once and for all. Instead, it sharpens the late-universe side of the dispute by reporting a local expansion rate of **$H_0 = 73.50 \pm 0.81 \text{ km/s/Mpc}$** , reaching roughly 1% precision through a unified network of multiple distance indicators rather than a single calibration chain.

At the same time, the broader field is still divided. Wendy Freedman's 2025 JWST-led calibration work argued there was no strong evidence for a severe Hubble tension, so the current state of play is better described as unresolved than solved.

RUT Interpretation

If the Hubble tension reflects genuine late-time structure or transition behavior rather than a pure calibration artifact, then improved low-redshift measurements should continue to prefer either a persistent positive offset in inferred H_0 or a repeatable redshift-dependent residual pattern relative to baseline Λ CDM expectations.

***RUT-style prediction:** independent low- z datasets should continue to favor a late-time H_0 above the standard early-universe inference if the late-time RUT sector is approximately correct.*

Predictions

- The discrepancy should not wash out purely as random scatter; it should appear as a stable offset, a structured redshift-dependent residual signature, or both.
- As covariance-aware local networks improve, a high late-time H_0 should remain difficult to erase if the late-time RUT picture is approximately right.
- If future datasets decisively collapse toward the early-universe value, the late-time RUT interpretation weakens substantially.

Falsifiers

- Future local-network analyses converge toward the early-universe Λ CDM value.
- Residual structure disappears once systematics are tightened.
- Joint fits across supernovae, BAO, cosmic chronometers, growth, and lensing fail to justify the extra structure of the RUT model.

- Independent recalibration programs continue reducing the discrepancy without requiring new late-time physics.

Reference Note

Sources referenced in the text: ISSI Bern press release on the 2026 H0 Distance Network result; University of Chicago news summary of the 2025 Wendy Freedman JWST-based calibration result.